



Fast-track Your Wireless Sensing Application Development Using MCXW71 and Low-Power Magnetic Wakeup Sensors

Hands-on Lab#2 Guide

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Hands-On Lab#2 Overview:

Tamper detection using low-power magnetic wakeup sensor:
Securing Your Assets with MCXW71 Wireless MCU

Brief Description

This example uses **FRDM-MCXW71** and **NMH1000** magnetic switch sensor (using sensor expansion board) to demonstrate autonomous detection of tampering/theft/abuse on device using low-power magnetic wakeup feature and transmit ALERT message via BLE wireless UART.

Details

The core idea is to leverage advanced sensor technology to protect high-value or secure assets. For example, smart meters can monitor for tampering, enhancing home security by detecting unauthorized access to safes or lockers. Personal medical devices can be safeguarded against misuse, and laptops or tablets can be protected from theft. In industrial settings, warehouse theft detection and machine tampering alerts can significantly reduce losses and improve security. This technology can benefit even simple applications like door open/close detection.

Solution

This example demonstrates ease-of-enablement using NXP's FRDM-MCXW71 and sensors development ecosystem to accelerate prototyping for your multiple such applications.

The **Lab#2** solution comprises of

- **MCX W71** for transmitting tamper detection alerts via BLE.
- Hall effect magnetic switch sensor **NMH1000** for low-power magnetic wakeup.

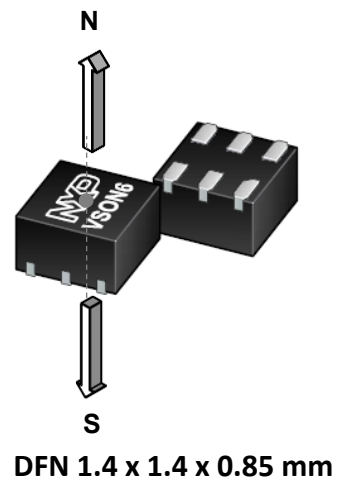
Recommended Sensor

NMH1000: Hall-Effect Magnetic Switch

The **NMH1000** is a **Hall effect magnetic field switch** with small physical size. The switch is most sensitive to **a vertical field passing through** the top-to-bottom surfaces of the package. The switch operates at low supply voltage with **ultra-low power consumption**. The switch detects the absence of a magnetic field and outputs either High or Low state upon the comparison with pre-set thresholds.

Features:

- Z- axis Hall effect switch for axial magnets
North & South polarity sensitivity
- Low operating voltage:
1.2 V min
- Ultra low current:
75 nA typ. @ 1 Hz sample rate
- Industrial operating temperature range:
-40 °C to 85 °C
- Selections by hardware pin / register:
 - Threshold
 - Sample Rate
 - *Standalone Mode or I²C User Mode*
- 2-state output via hardware pin / register

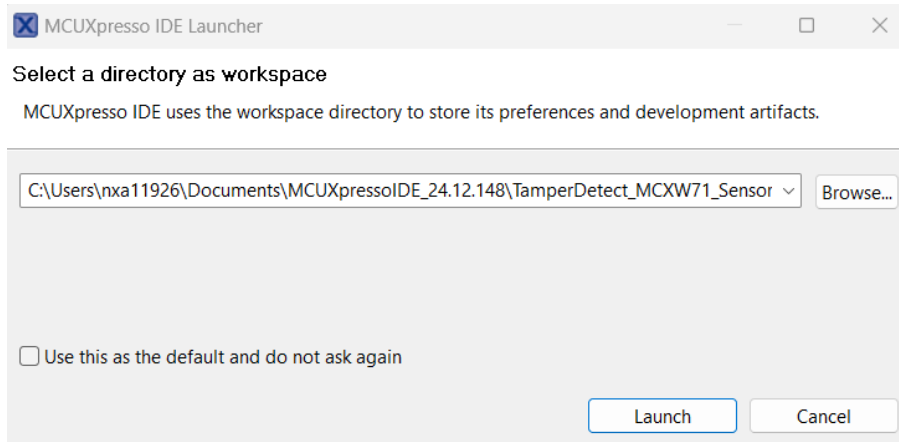


Target Applications

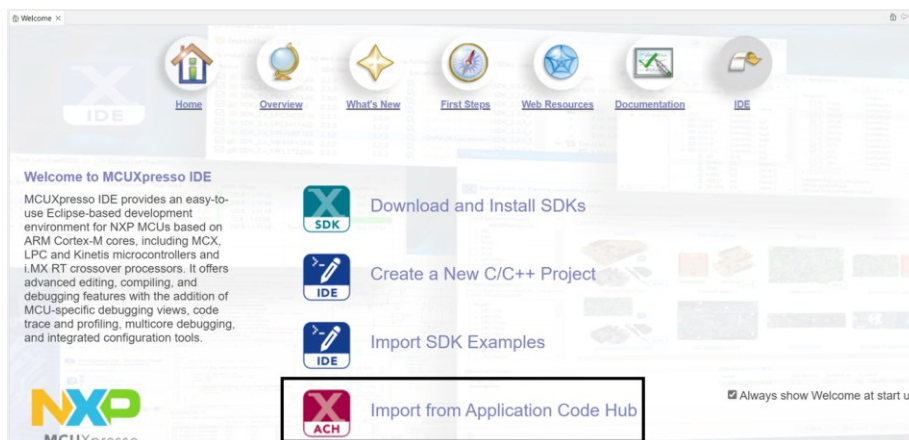
- Door or window open / closed detection
- Laptop lid open / closed detection
- IOT applications
- Electronic system wake-up
- Drug delivery device
- CGM application
- Insulin pump flow control
- Metering anti-tampering
- Reed switch replacement

Get Hands-on Workshop Lab Examples from ACH

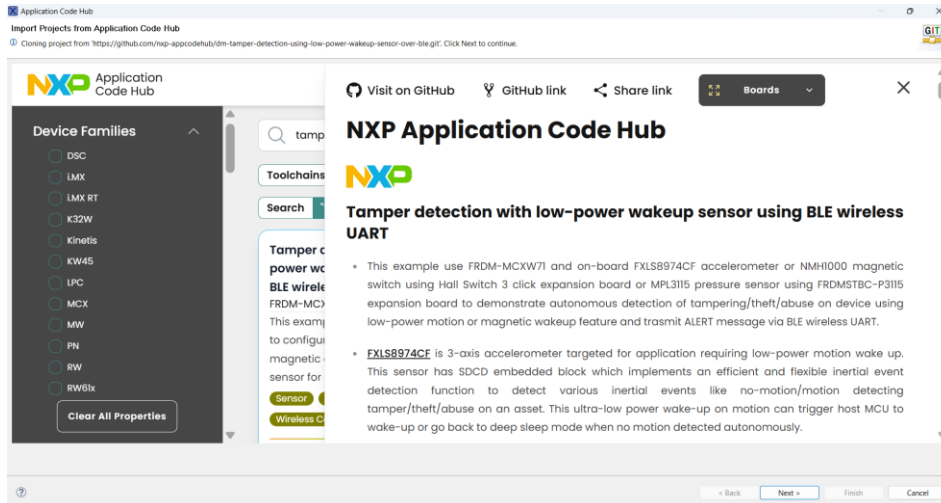
- 1) Launch MCUXpresso IDE and create workspace name as “TamperDetect_MCXW71_Sensors”.



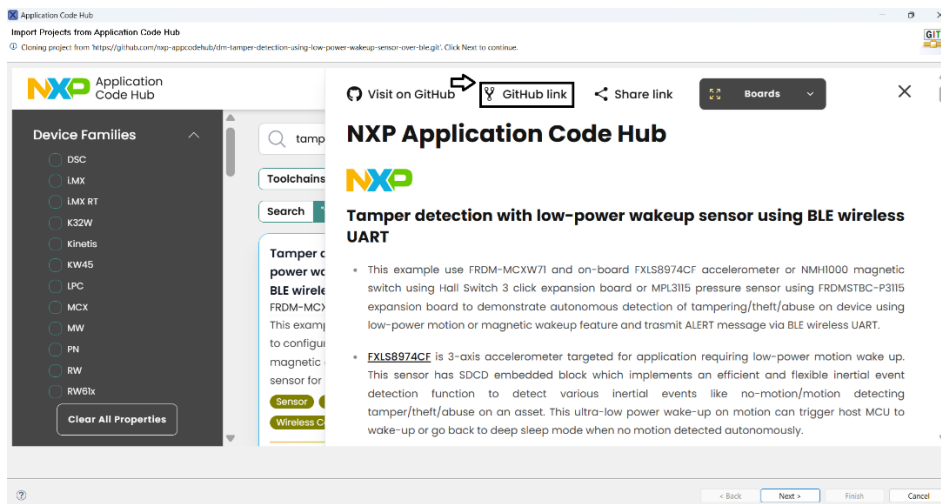
- 2) Select “Import from Application Code Hub” on the welcome screen.



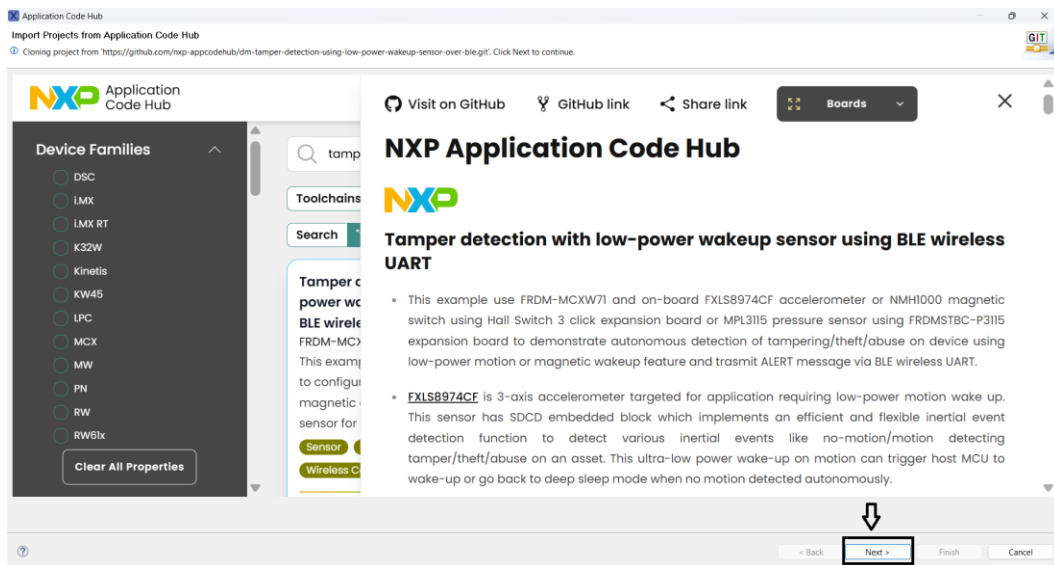
- 3) It will open a view, type in “Tamper detection FRDM-MCXW” in the search window. Select the “**Tamper Detection with low-power sensors using BLE wireless UART**” example as shown in the snapshot.



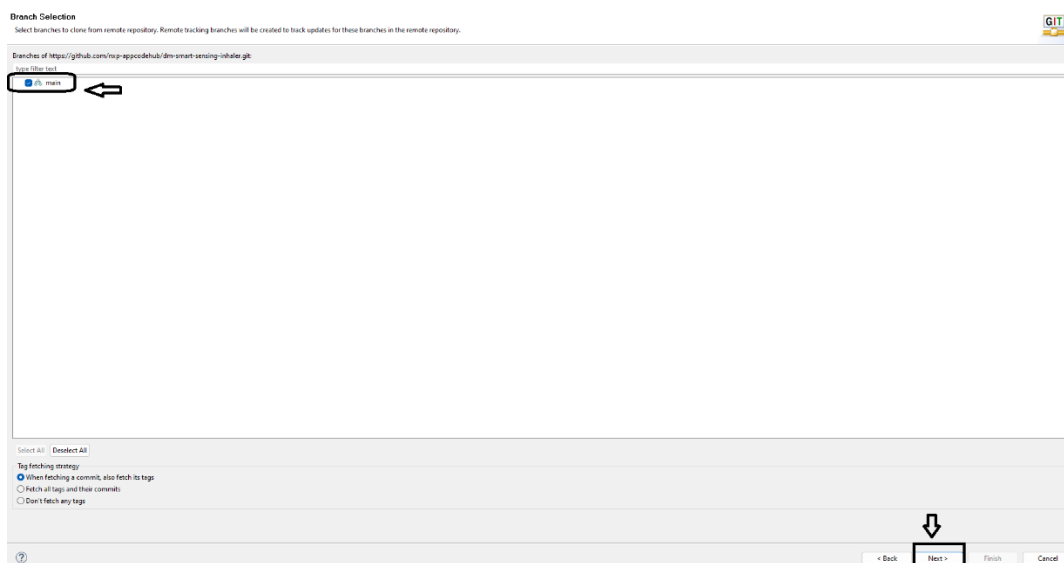
4) Click on “GitHub link”, it will start cloning the chosen example project into workspace folder.



5) The chosen example project is now getting imported into MCUXpresso IDE workspace. Click “Next”



6) Select git branch as “main” and click “Next”.



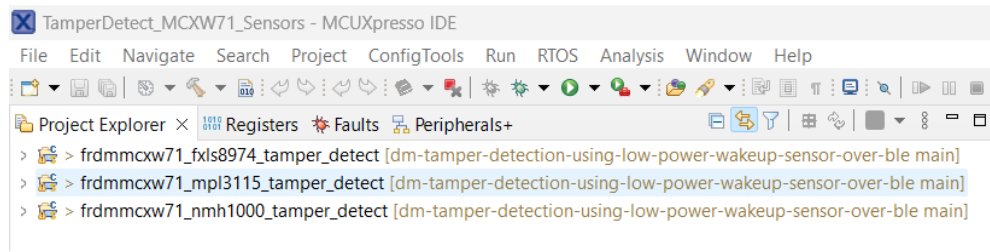
7) Select the local storage locate for cloned project. By default, the tool will select C:\Users\<userid>\git\ folder. Click “Next”.



8) Select projects available under “C:\Users\<userid>\git\dm-tamper-detection-using-low-power-wakeup-sensor-over-ble\tamper_detection_demo”. Click “Next”.

Note: Ignore any “Project SDK management” warning showing older version of SDK. Click on ok button (for all the 3 examples) and continue.

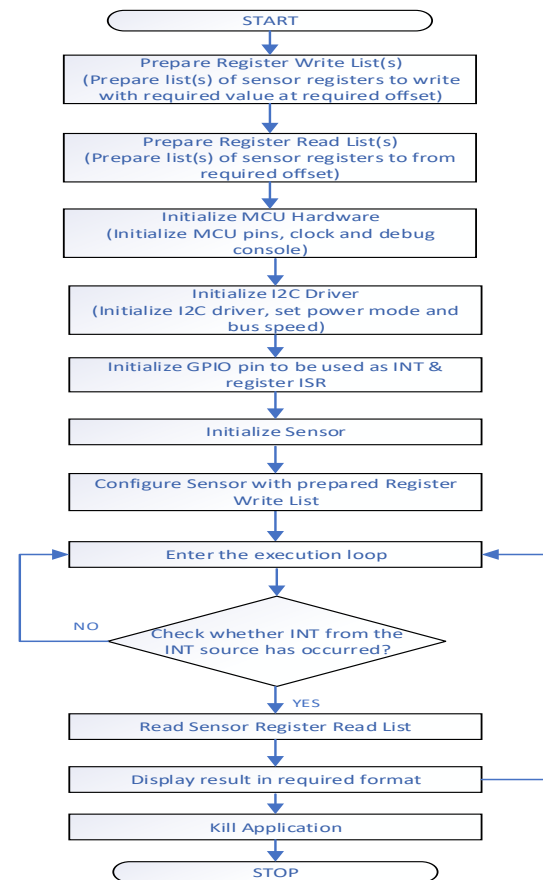
9) Click “Next”. MCUXpresso IDE will open the chosen projects.



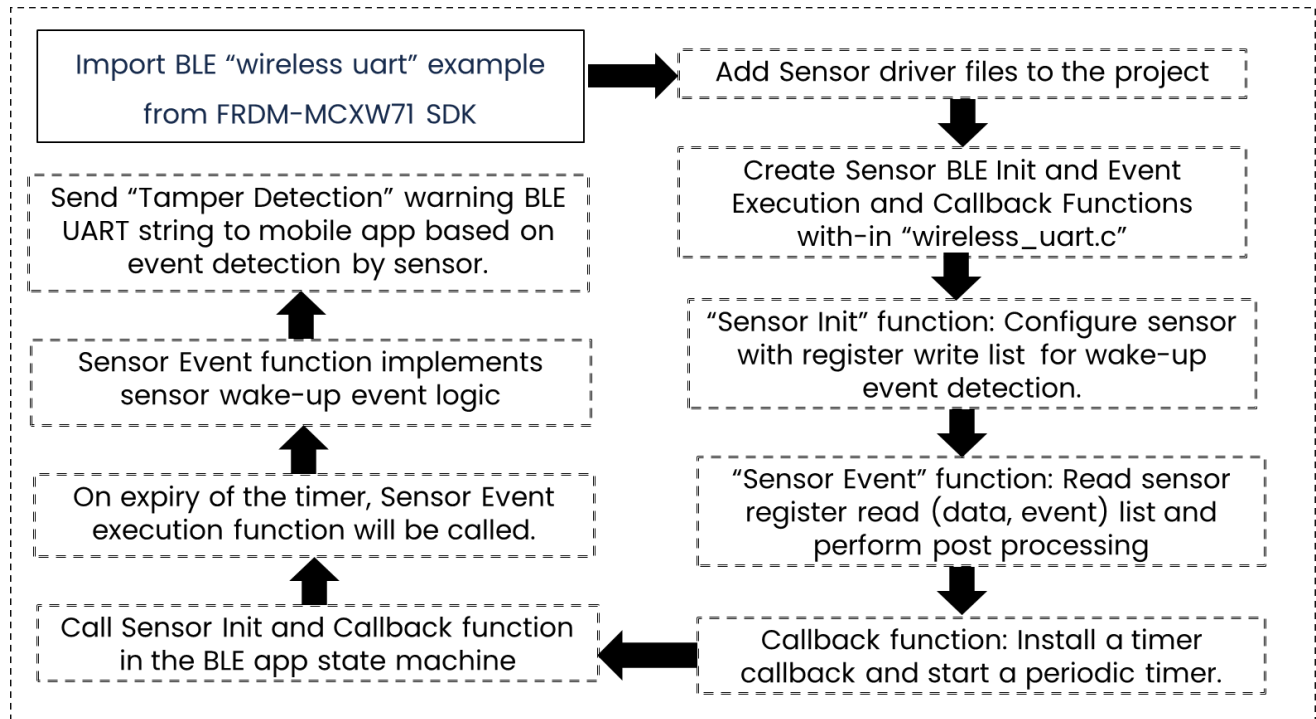
Code Walkthrough

Typical ISSDK example code flow:

- 1) Prepare register write list.
- 2) Prepare register read list.
- 3) Initialize MCU Hardware (Pins, Clock etc.)
- 4) Initialize COMM (I2C, SPI) driver
- 5) Initialize GPIO pin to be user as INT and register ISR.
- 6) Initialize Sensor, read WHOAMI
- 7) Configure Sensor with prepared register write list
- 8) Enter execution loop:
 - i. Check whether interrupt source has occurred.
 - ii. If yes, then read register read list
 - iii. Display result or upload into TSA table
 - iv. Execute post processing or other functions.



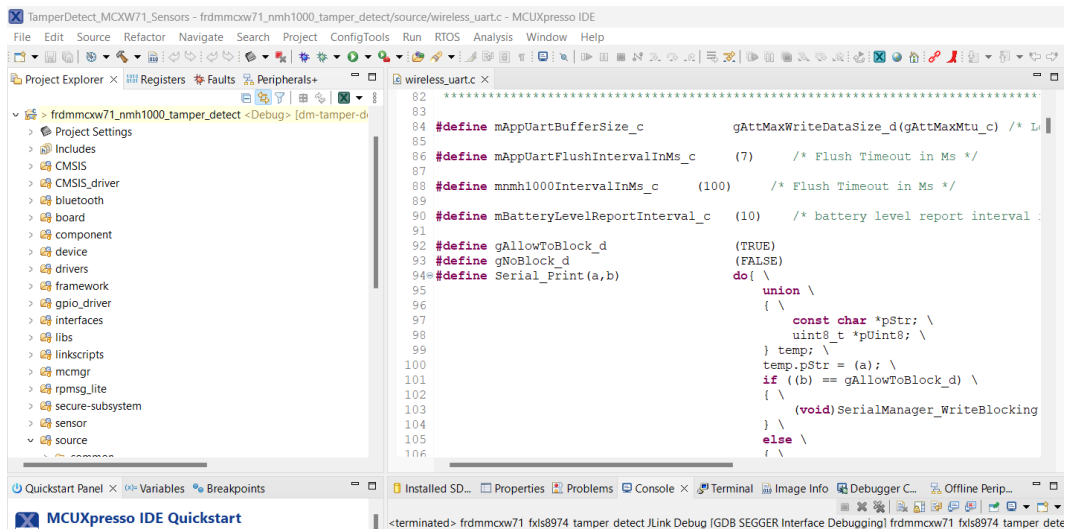
Learn how to convert SDK wireless example to wireless sensing example:



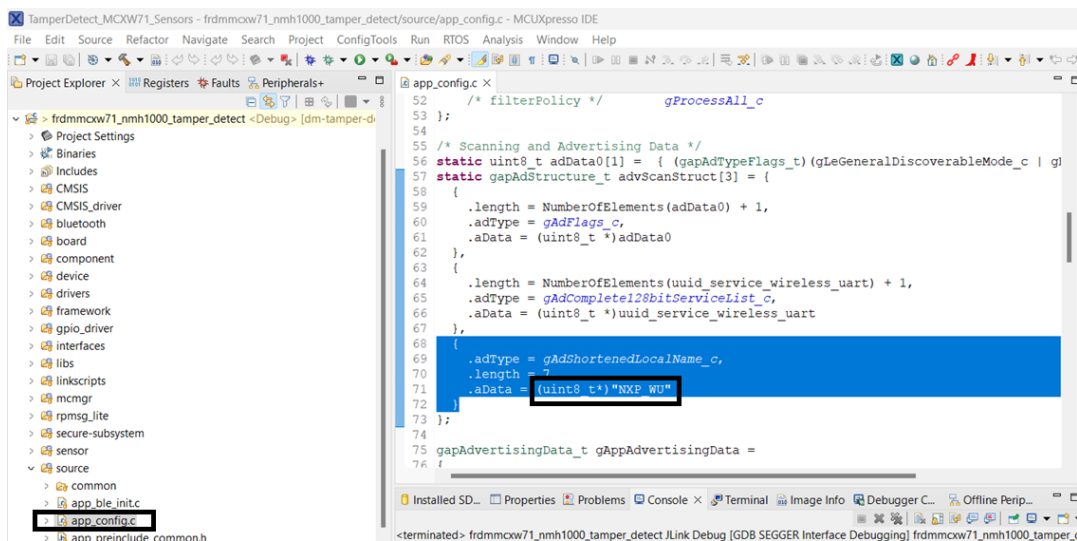
Hands-on Lab#2: Using Magnetic Switch Sensor

Compile Project:

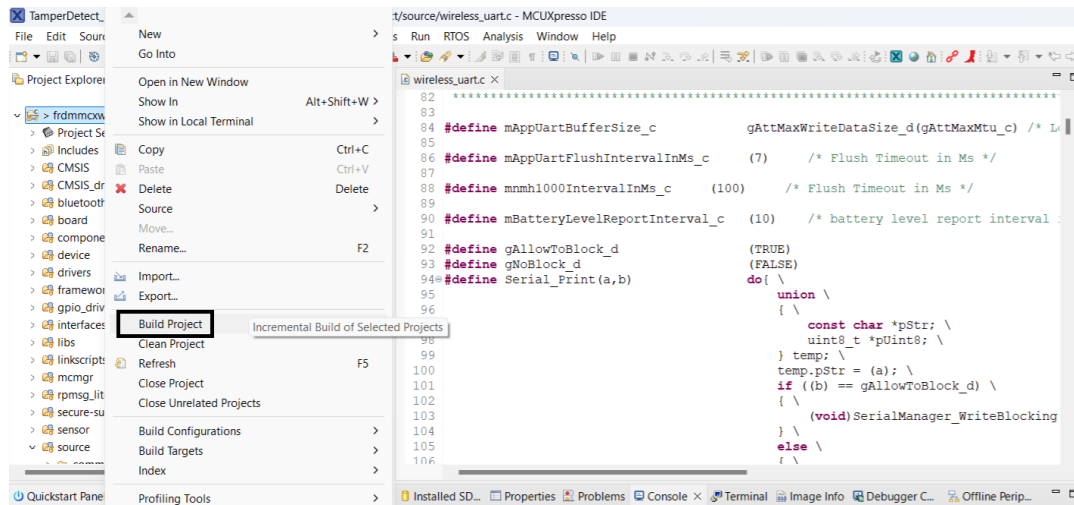
- 1) Chose “frdmmcxw71_nmh1000_tamper_detect” project” on your MCUXpresso IDE workspace “TamperDetect_MCXW71_Sensors”:



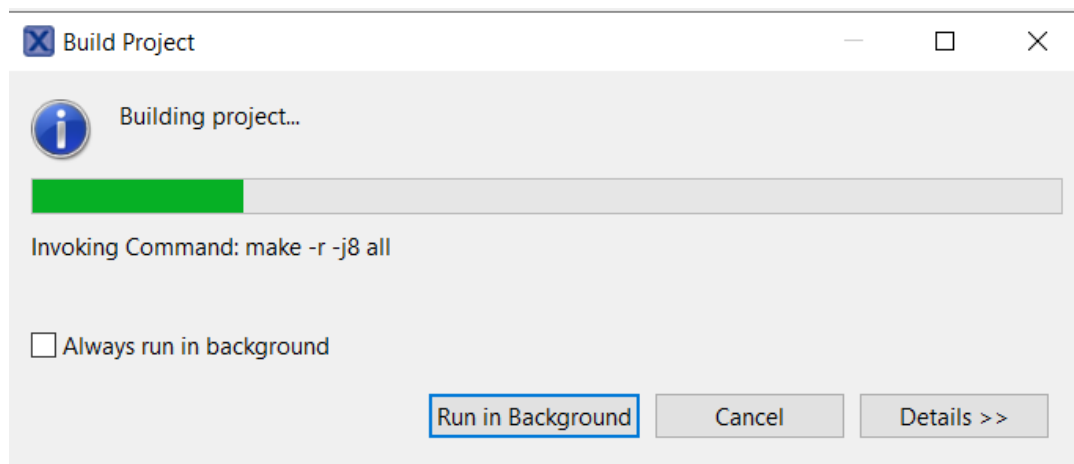
- 2) Before to compile the project go to Explorer view and open app_config.c file.
- 3) On line 71, make sure to update “aData” field with a unique name (this will help to identify your board so that it doesn’t clash with any other nearby board). Let me change to “NXP_M0” (try using NXPM01 onwards) but you can select any name you want (max length is 6 characters).



- 4) Right click on “frdmmcxw71_nmh1000_tamper_detect” project and select “Build Project”

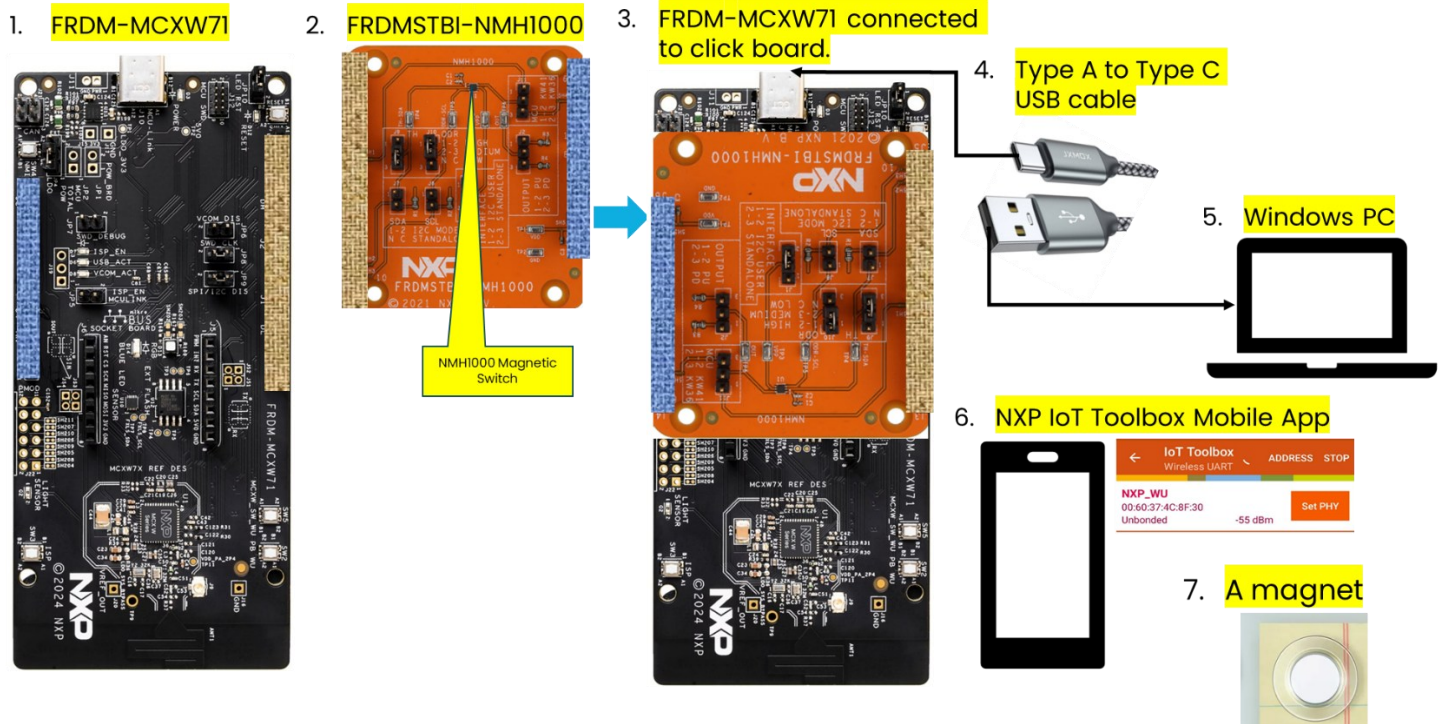


- 5) MCUXpresso IDE will start compiling the “frdmmcxw71_NMH1000_tamper_detect” project.



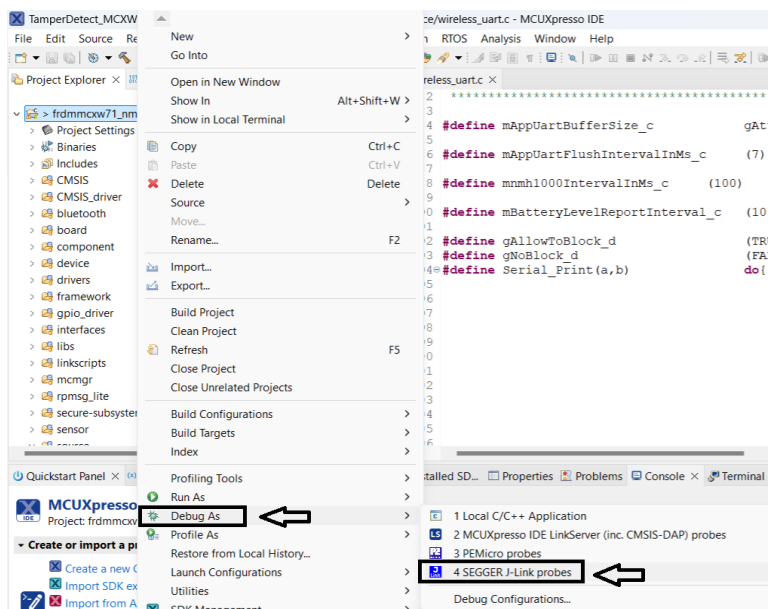
- 6) Confirm successful compilation of “frdmmcxw71_NMH1000_tamper_detect” project.

Connect HW:

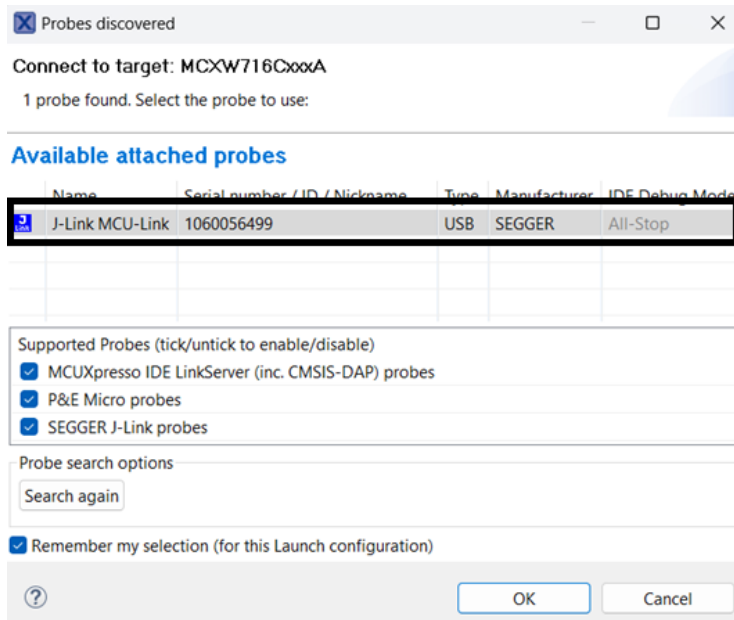


Program the target Board:

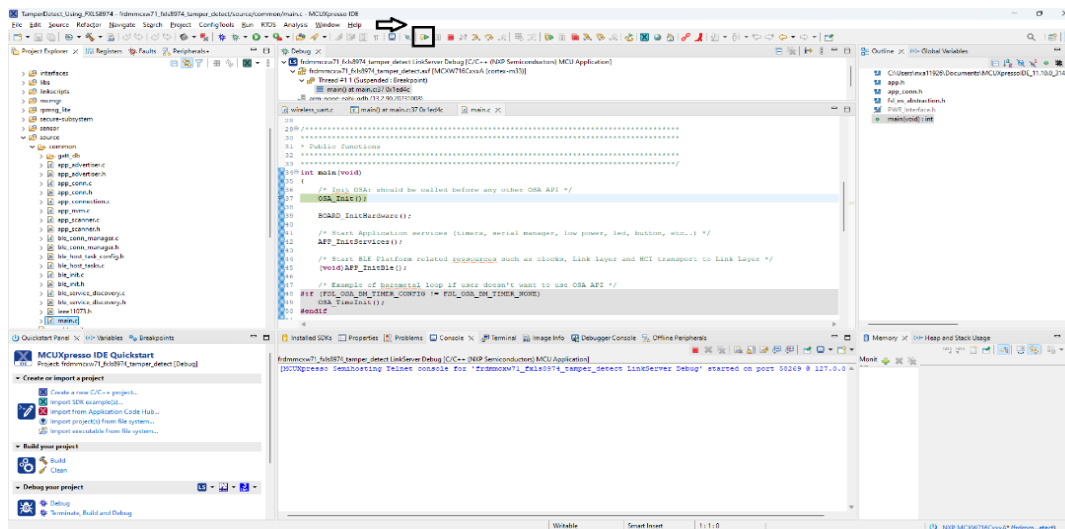
- 1) Right click on the project and select "Debug As", click on "SEGGER J-Link probes".



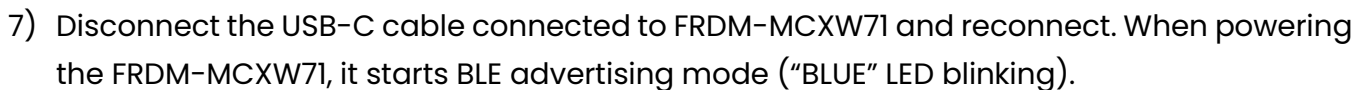
- 2) Debugger starts downloading the program to device. “J-Link MCU-LINK” probes will be identified. Click “OK” to continue.



- 3) Accept the J-Link – Terms of Use. Debugger will load the program to the target board.
- 4) Click on “Resume” button or press “F8” key on your keyboard to continue running the downloaded program on device.

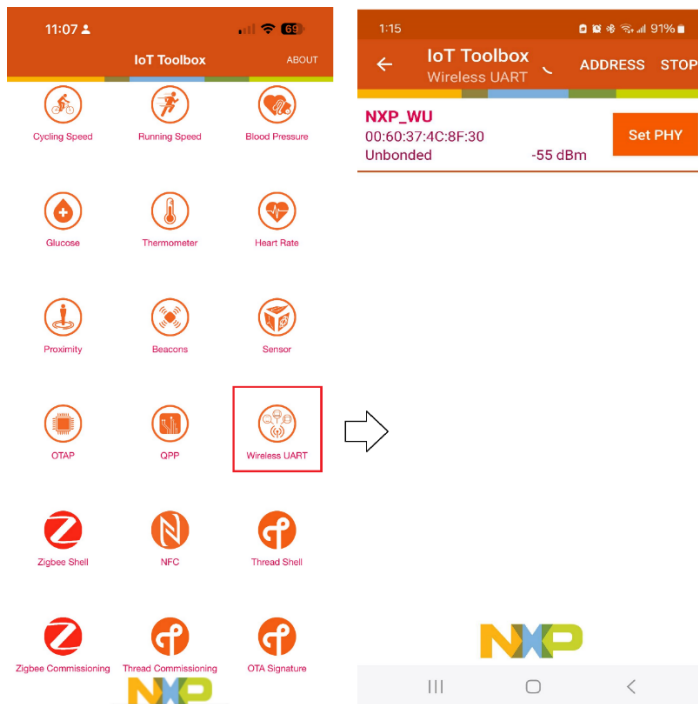


- 5) Now the program is downloaded on the device. you should see “BLUE” LED blinking.
- 6) Now the program is downloaded on the device. Click on “Terminate” button or press “CNTR + F2” to terminate the debug session.

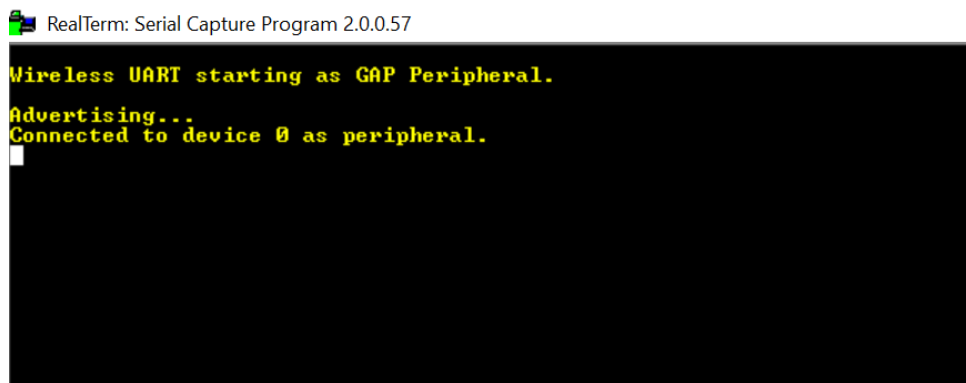


Run NXP IoT Toolbox App on your mobile:

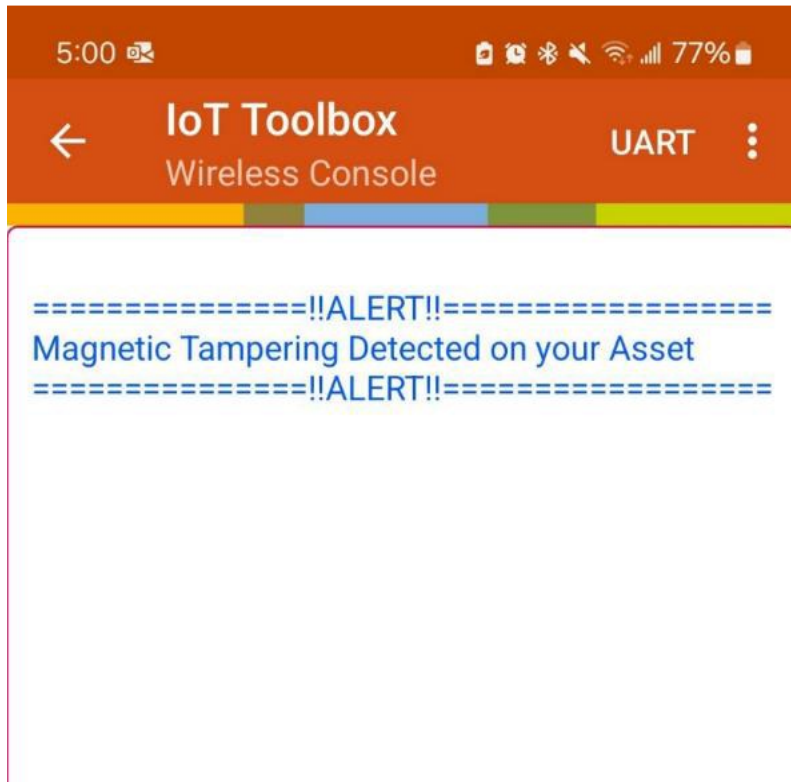
- 1) Open the installed NXP IoT Toolbox (available from Play Store and App Store) on your cell phone. Make sure Bluetooth connection is ON.
- 2) Inside the application, click on the Wireless UART widget. Scan should start automatically.
- 3) When a device called NXP_M0 or <your_id> appears, click on it. Your phone should now be connecting to the board.



- 4) After connecting with the FRDM-MCXW71 device, the serial UART terminal will show this message:



- 5) When you show magnetic tampering (put a magnet near to the NMH1000 on FRDMSTBI-NMH1000 connected to FRDM-MCXW71), the NMH1000 magnetic switch sensor detects change in magnetic field and wakeup.
- 6) At that point, you will see Wireless UART app showing ALERT message as shown below:



- 7) FRDM-MCXW71 board will also show "RED" LED status. The "RED" LED status will continuously remain ON till magnetic tampering is detected.

Congratulation on successfully completing Hands-On Lab2



Congratulations!
You successfully
completed the
Hands-On Lab#2